

Day 8 — Project Session 1: Plan and First Artefact

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will produce a one-paragraph project plan with element (lunar topic), format, audience, and citation; and complete a first-draft artefact (poster sketch / video storyboard / slide outline).

Materials

- Project supplies for whatever format students chose: poster paper, markers, A4 sheets, laptops, art supplies
- Printed copy of the rubric (`assessments/rubric.md`) — one per pair
- The class *pañcāṅga* (for reference)
- Days 1–7 student notebooks (their own diaries + worksheets)

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- Hand each pair a printed rubric. “*Read through the rubric — 4 minutes. Bring questions to me.*”

Core (5 min)

Take questions on the rubric. The most common questions:

- “*Do we need to quote a verse?*” — No. A paraphrase of the *Sūrya-siddhānta* (e.g. “the *Sūrya-siddhānta* defines a tithi as 12° of elongation”) is enough. A NASA reference is also acceptable. The rubric cares that your source is real and named.
- “*Can we do TWO topics?*” — One main topic. You may compare to a second briefly.
- “*What if our diary is missing some nights (clouds)?*” — Note “cloudy, predicted [shape]” for those nights. The diary still counts.

Activity (30 min) — Build

Pairs work on their projects. The teacher rotates and checks each pair has:

1. **Chosen topic** — one of:
 - Moon phases (geometry and the 10-night diary visualised)
 - Eclipses (why, when, safety)
 - Tithi calendar (the one built on Day 4 + an explanation)
 - Cross-cultural lunar calendars (preview of Day 9)
 - Lunar months and festivals

- Adhika māsa (intercalary month)
- 2. **Chosen format** — one of:
 - Poster (A2 or larger)
 - Slide deck (5–8 slides, projected)
 - 90-second video
 - 1-page printed story / explainer
 - Live demonstration with the torch and ball
 - Infographic / data visualisation
- 3. **One source named** — *Sūrya-siddhānta* (paraphrased), NCERT Physics XI, NASA’s phase calendar, the class *pañcāṅga*, etc. **Real sources only.**
- 4. **By end of class:** a one-paragraph plan + a first-draft artefact (a poster sketch, video storyboard, slide outline).

The teacher intervenes if: - A pair is still debating topic at minute 15 — give them 5 minutes, then assign. - A pair’s source is invented or unverifiable — flag it; help them swap. - A pair has split unequally — re-balance the workload before they leave.

Wrap-up (5 min)

- Each pair submits their plan paragraph + first-draft artefact (phone photo accepted).
- **Tomorrow:** finish the build. The cross-cultural calendar lesson runs in parallel — 15 min content, then 30 min build time. Be ready.

Student-facing content

Today is build day.

By the end of this class, your pair (or you, if solo) should have: - A one-paragraph plan in writing. - A first-draft artefact — poster sketch, slide outline, video storyboard. - Your source named, on the artefact.

Tomorrow you refine. Day 10 you present.

Homework

- Continue the Moon diary (you have 2 nights left).
- Finish at home what you couldn’t finish in class. Don’t *start* something new — refine.

Differentiation

- **One tier up:** Add a second citation. Two sources is a stronger project — for instance, paraphrase the *Sūrya-siddhānta* AND cite NASA’s phase calendar.
- **One tier down:** Reduce scope. Pick ONE phenomenon (phases, eclipses, or the calendar) and explain it in one paragraph + one diagram. That’s a complete project.

Teacher notes

- Pairs that are still “researching” by minute 20 are stuck. Push them to start sketching/drafting even if incomplete. The artefact concretises the thinking.

- Phone-photo submissions are fine. Photograph the draft poster against a wall; submit by email or via shared drive.
- A common failure mode: pairs over-design the visual and under-write the explanation. Push toward writing the **3-minute script** by end of class.

Citation(s) used in this lesson

- The rubric (`assessments/rubric.md`) is the primary reference for what “good” looks like.
- Source-quality reminder: ONLY real sources. No invented citations.